

Offset-Aware Bidirectional Spatio-Temporal Aggregation for Remote Sensing Video Super-Resolution

Senwei Hu, Yisheng Zhao, Member, IEEE, Hongyi Chen, Yaoqi Liu, Member, IEEE, Yibo Wang, Hao Zhou, Yinhe Han, Senior Member, IEEE, Jianyu Wang

Abstract—High-resolution satellite videos are essential for video-based remote sensing analysis, yet their acquisition and transmission are constrained by sensor bandwidth, onboard storage, and downlink capacity. Remote sensing video super-resolution (RSVSR) aims to recover high-resolution video sequences from low-resolution satellite observations. Compared with natural videos, satellite videos pose distinctive challenges due to platform-induced motion, weak local textures, and dense structure-sensitive details. Existing RSVSR methods are often evaluated on datasets with limited temporal variation, whereas realistic satellite videos exhibit stronger frame-to-frame changes. To better reflect such temporal conditions, we construct CQ1-VSR from original satellite videos without precise inter-frame registration. The HR source sequences are real satellite observations, and the paired LR inputs are generated through a documented clip-consistent degradation pipeline. We propose BiSTAVSR, a recurrent video restoration framework that enhances local spatio-temporal aggregation within long-range propagation. It incorporates bidirectional local attention, structure-guided offset refinement, and an explicit high-frequency enhancement branch to improve spatio-temporal feature utilization and detail reconstruction. Experiments on SAT-MTB-VSR and CQ1-VSR demonstrate that BiSTAVSR achieves favorable quantitative performance and visual quality, particularly in regions with complex motion and fine structural details. Code, models, degradation scripts, and dataset are available at <https://github.com/HIAS-Zhao/Video-super-resolution>.

Index Terms—Satellite video, video super-resolution, spatio-temporal aggregation, offset-aware refinement, high-frequency reconstruction.

Copyright © 20xx IEEE. Personal use of this material is permitted. However, permission to use this material for any other purposes must be obtained from the IEEE by sending an email to pubs-permissions@ieee.org.

This work was supported in part by the National Key Research and Development Program of China (No. 2022YFB3902802), the National Natural Science Foundation of China (No. 62025404), and the Beijing Natural Science Foundation (No. L241013, L258047). (Corresponding author: Yisheng Zhao.)

Senwei Hu, Yisheng Zhao, Hongyi Chen, Yibo Wang, Hao Zhou are with the Hangzhou Institute for Advanced Study, University of Chinese Academy of Sciences, Hangzhou 310024, China (e-mail: husenwei25@ucas.ac.cn, wangyibo251@ucas.ac.cn, zhouhao24@mails.ucas.ac.cn, zhaoyisheng@ucas.ac.cn).

Yaoqi Liu, Yinhe Han are with the State Key Lab of Processors, Institute of Computing Technology, Chinese Academy of Sciences, Beijing 100089, China (e-mail: liuyaoqi@ict.ac.cn, yinhes@ict.ac.cn).

Jianyu Wang is with the Hangzhou Institute for Advanced Study, University of Chinese Academy of Sciences, Hangzhou 310024, China; University of Chinese Academy of Sciences, Beijing 100049, China; Shanghai Institute of Technical Physics, Chinese Academy of Sciences, Shanghai 200083, China (e-mail: jywang@mail.sitp.ac.cn).

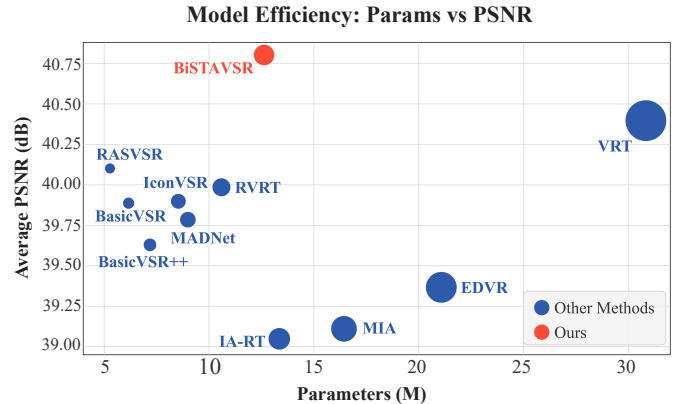


Fig. 1: Parameter-PSNR comparison on the SAT-MTB-VSR evaluation split.

I. Introduction

SATELLITE videos provide high-temporal-resolution observations of the Earth’s surface and are increasingly used in video-based remote sensing applications, including urban monitoring, maritime surveillance, vehicle tracking, event understanding, and fine-grained interpretation of man-made structures [1]–[3]. Compared with still remote sensing imagery, satellite video sequences contain dynamic scene information and inter-frame temporal cues that are valuable for both downstream analysis and spatial reconstruction. In practical satellite-to-ground pipelines, however, the transmitted video resolution is constrained by sensor bandwidth, onboard storage, downlink capacity, and real-time processing requirements. Remote sensing video super-resolution (RSVSR) is therefore useful for bandwidth-efficient video observation, where lower-resolution satellite videos can be transmitted and high-resolution video sequences are reconstructed on the ground.

RSVSR differs substantially from conventional video super-resolution on natural videos. Satellite videos are captured from a high-altitude moving platform, so adjacent frames often contain scene-scale translation, view-point changes, imaging-geometry variation, and parallax. At the same time, remote sensing scenes usually have weak local texture but dense structure-sensitive details, such as roads, building boundaries, shorelines, ports, and small

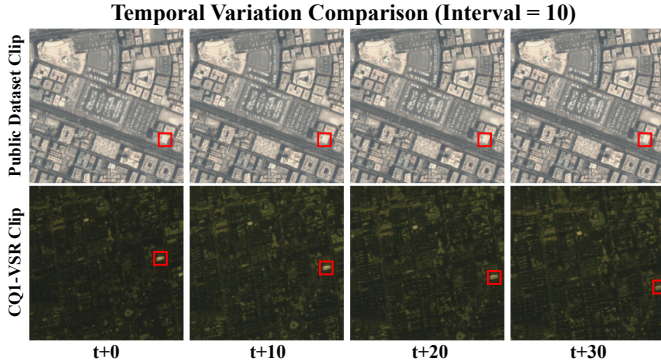


Fig. 2: Temporal variation comparison between a public RSVSR benchmark clip and a CQ1-VSR clip. Frames are sampled every 10 time steps. Compared with the public dataset clip, CQ1-VSR shows larger scene displacement and stronger inter-frame variation, making temporal aggregation more challenging.

vehicles. These properties make spatio-temporal feature fusion highly sensitive to alignment errors: an inaccurate correspondence can duplicate edges, blur small structures, or introduce ghosting artifacts across the restored video sequence. Realistic RSVSR therefore requires not only coarse motion guidance, but also structure-aware local matching and conservative high-frequency reconstruction. This requirement is particularly important when the LR observation is ambiguous: useful details may exist in neighboring frames, but they can only be exploited if the model can distinguish reliable spatio-temporal support from misaligned or temporally inconsistent evidence.

Existing RSVSR benchmarks, such as JILIN-1 [4] and SAT-MTB-VSR [5], have played an important role in standardizing evaluation. However, many sequences in these benchmarks are well aligned or processed under relatively controlled conditions, which can reduce frame-to-frame variation. To complement them, we construct CQ1-VSR from original QK-1 satellite videos. CQ1-VSR retains stronger platform-induced inter-frame variation, as visually illustrated in Fig. 2, and provides a more realistic testbed for evaluating whether a model can exploit neighboring frames under larger temporal displacement.

Existing VSR architectures are not fully matched to this satellite video setting. Recurrent methods [6]–[11] scale well to long video sequences, but they compress long-range history into a hidden state that must also absorb current-frame evidence and compensate for alignment errors. When neighboring frames differ strongly, this hidden state may dilute useful local evidence around the target frame, especially for thin structures that appear only briefly in the local temporal neighborhood. Sliding-window and transformer-based methods [12]–[15] directly aggregate neighboring frames and often provide stronger target-centered support, but their cost grows with the number of input frames and their local fusion may mix past and future evidence before sufficient directional reasoning. Moreover, standard reconstruction heads tend to smooth

high-frequency structures that are important in remote sensing video interpretation, such as building contours and road boundaries.

Motivated by these observations, we propose BiSTAVSR, an offset-aware bidirectional spatio-temporal aggregation framework for realistic RSVSR. The framework preserves recurrent propagation for efficient long-range video modeling, while inserting an explicit local aggregation module around each target frame. BiSTA separately aggregates past and future local support with deformable attention and injects the two results into the current recurrent state, preserving temporal directionality instead of mixing all support frames at once. OAR provides structure-aware guidance for offset prediction and query-key matching, so offset estimation is encouraged to follow stable boundaries rather than noisy texture responses. PHASE adds an explicit high-frequency residual branch for detail compensation, allowing the main reconstruction path to remain stable while the auxiliary path sharpens reliable structures. As shown in Fig. 1, the proposed design achieves a favorable balance between video reconstruction quality and parameter efficiency.

The main contributions of this work are summarized as follows:

- CQ1-VSR is introduced as a satellite video super-resolution dataset constructed from original satellite videos with synthetic LR-HR supervision. It complements existing RSVSR benchmarks by retaining larger inter-frame variation and richer platform-induced motion in the HR video sequences.
- We propose BiSTAVSR, a recurrent RSVSR framework that strengthens local neighboring-frame aggregation while preserving efficient long-range bidirectional video propagation.
- Three task-oriented modules are developed for realistic satellite video restoration: BiSTA for symmetric local spatio-temporal aggregation, OAR for structure-guided offset and query-key refinement, and PHASE for explicit high-frequency detail compensation.
- We conduct experiments on SAT-MTB-VSR and CQ1-VSR, including SOTA comparison, temporal consistency evaluation, ablations, visual analysis, and efficiency comparison, to assess both frame-level reconstruction and video-level temporal behavior.

II. Related Work

A. Video Super-Resolution

Video super-resolution reconstructs high-resolution video sequences by exploiting spatial detail and temporal redundancy. Mainstream VSR methods differ mainly in how they align, propagate, and aggregate neighboring frames. Window-based and multi-branch methods aggregate local neighboring frames for target-frame reconstruction [12], [16], while recurrent methods such as BasicVSR [6], BasicVSR++ [7], and RVRT [8] improve scalability by propagating features over time. Transformer

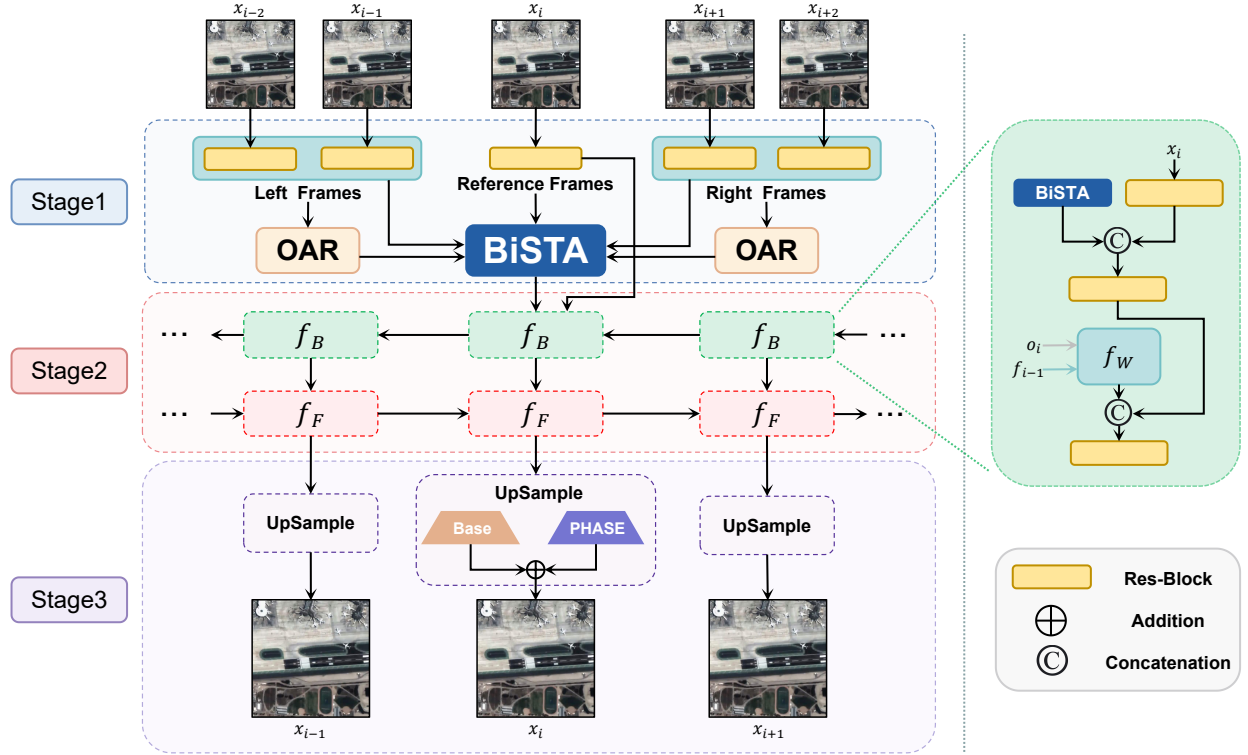


Fig. 3: Overall architecture of BiSTAVSR. The framework first extracts shallow frame features and estimates bidirectional optical flow, and then performs OAR-guided local BiSTA aggregation, bidirectional recurrent propagation, and dual-branch reconstruction with explicit high-frequency enhancement.

and resampling-based methods such as VRT [13], MIA-VSR [14], and IA-RT [15] further strengthen long-range aggregation or alignment. These methods motivate different tradeoffs between local evidence, long-range propagation, and alignment flexibility. BiSTAVSR follows the recurrent paradigm but augments each propagation step with direct local bidirectional support for satellite videos with scene-scale motion.

B. Remote Sensing Super-Resolution

Remote sensing imagery contains dense repetitive structures, weak local texture, and overhead viewpoints, making direct transfer from natural-image restoration difficult. Early satellite video SR studies introduced CNN-based reconstruction and progressively enhanced fusion for JILIN-1 videos [17]–[19]. Later methods improved satellite video restoration by exploiting adaptive spatio-temporal neighbors, modeling multiple degradations, supporting arbitrary-scale reconstruction, and learning from a single video without external supervision [20]–[23]. More recent RSVSR methods further strengthen temporal modeling through local-global temporal differences, non-local bidirectional recurrence, lightweight propagation, continuous-scale restoration, and multi-axis feature enhancement [9]–[11], [24], [25]. These studies highlight that satellite video restoration requires domain-specific spatio-temporal modeling rather than direct transfer from natural video SR. Complementary to these works, BiSTAVSR focuses on explicit local past/future aggregation

under stronger real satellite-video motion, while CQ1-VSR provides a benchmark for evaluating this setting.

C. Alignment and Deformable Attention

Temporal alignment is central to VSR because useful information is distributed across multiple neighboring frames. Optical-flow warping provides explicit motion estimates [26], [27], and flow-based recurrence has also been explored for efficient long-range video restoration [28]. Nevertheless, flow errors are common in low-texture regions, occlusions, and complex parallax. Deformable convolution and deformable attention mitigate this issue through adaptive sampling [29], as shown by EDVR [12], BasicVSR++ [7], RVRT [8], and deformable-attention-based models [30]. Recent implicit resampling also revisits the alignment problem from a continuous-coordinate perspective [15]. In remote sensing videos, however, inaccurate offsets can easily duplicate or blur thin structures such as roads, roofs, and shorelines across the reconstructed sequence. BiSTAVSR therefore combines coarse flow guidance with learnable residual offsets, and uses OAR-refined query-key features to make offset prediction more structure-aware while preserving raw value features.

D. High-Frequency and Structure-Aware Restoration

High-frequency restoration is important for both perceptual quality and remote sensing interpretation. Frequency and channel attention have been used to emphasize informative responses in restoration and recognition

networks [31]–[33], and recent VSR work also explores multi-frequency representations for detail recovery [34]. Related remote sensing work has studied edge-aware losses, frequency-aware representations [35], and auxiliary enhancement branches [36]. In RSVSR, high-frequency details often correspond to building outlines, narrow roads, ship contours, or small vehicles. PHASE follows this motivation by combining adaptive feature enhancement, fixed structural priors, and directional responses to produce an explicit high-frequency residual.

E. Remote Sensing Video Datasets

Datasets determine what type of video temporal reasoning is evaluated. JILIN-1 [4] and SAT-MTB-VSR [5] are valuable standardized benchmarks, but many sequences are well aligned or processed in ways that reduce frame-to-frame variation. CQ1-VSR complements these benchmarks by retaining original platform-induced motion from QK-1 satellite videos, thereby providing a more suitable setting for examining local temporal aggregation, offset-guided alignment, and structure-aware video reconstruction under stronger inter-frame variation.

III. Method

A. Problem Formulation

Let $\mathbf{I}^{LR} = \{I_1^{LR}, I_2^{LR}, \dots, I_T^{LR}\}$ denote an LR video, where $I_t^{LR} \in \mathbb{R}^{C \times H \times W}$. For $\times s$ VSR, the goal is to estimate $\hat{\mathbf{I}}^{HR}$ with $\hat{I}_t^{HR} \in \mathbb{R}^{C \times sH \times sW}$; in this work, $s = 4$. For each target time step t , BiSTAVSR uses both a recurrent state H_t^{rec} and a local temporal neighborhood

$$\mathcal{N}_t^k = \{I_{t-k}^{LR}, \dots, I_{t-1}^{LR}, I_{t+1}^{LR}, \dots, I_{t+k}^{LR}\}, \quad (1)$$

where $k = 2$ by default. The recurrent state provides long-range memory, while $\mathcal{N}_t^k$ supplies direct target-centered evidence. The reconstruction function is formulated as

$$\hat{I}_t^{HR} = f_\theta(I_t^{LR}, \mathcal{N}_t^k, H_t^{rec}, \mathcal{F}_t), \quad (2)$$

where $f_\theta(\cdot)$ denotes the learnable RSVSR reconstruction function parameterized by θ , and $\mathcal{F}_t$ denotes optical-flow-based motion cues. The model is trained with the reconstruction loss

$$\mathcal{L}_{rec} = \frac{1}{T} \sum_{t=1}^T \rho(\hat{I}_t^{HR} - I_t^{HR}), \quad (3)$$

where $\rho(\cdot)$ is the Charbonnier penalty.

B. Overall Architecture

As shown in Fig. 3, shallow feature extraction and motion estimation first provide per-frame features and bidirectional optical flow. Built on MADNet, BiSTAVSR retains its MADE-based feature enhancement and EDVR keyframe branch, and organizes the subsequent processing into three stages: target-centered local BiSTA aggregation with OAR guidance, bidirectional recurrent propagation, and dual-branch reconstruction. The EDVR-based keyframe features are extracted at a fixed stride and

used as sparse stabilization anchors. During backward and forward propagation, each state is warped by flow and fused with the current-frame feature and the local BiSTA result. Non-OAR shallow support features are kept for value propagation, while OAR-refined features guide query-key matching and offset prediction.

The final propagated feature is decoded by a main branch and a PHASE detail branch. The main branch reconstructs base content, while the auxiliary branch predicts a high-frequency residual. The final output is

$$\hat{\mathbf{I}}_t^{HR} = \mathbf{I}_t^{main} + \alpha \mathbf{I}_t^{detail} + \mathbf{I}_t^{base}, \quad (4)$$

where $\mathbf{I}_t^{main}$ and $\mathbf{I}_t^{detail}$ denote the outputs of the main and auxiliary detail branches, respectively, $\mathbf{I}_t^{base}$ denotes the $\times 4$ bilinear interpolation of the input frame, and α is a learnable global scaling factor for detail compensation.

C. Bidirectional Spatio-Temporal Attention Aggregation

As illustrated in Fig. 4, BiSTA strengthens local multi-frame support inside recurrent propagation. For target frame t , let F_t be the current feature and H_t^{rec} be the propagated recurrent feature. We collect local support features $\mathcal{N}_t = \{F_{t-k}, \dots, F_{t-1}, F_{t+1}, \dots, F_{t+k}\}$ with $k = 2$ by default and align non-adjacent frames using composed flows. Unlike the recurrent hidden state, this local support is target-centered and short-range, providing complementary evidence when propagation becomes over-smoothed or affected by accumulated alignment errors. BiSTA separates $\mathcal{N}_t$ into past and future groups, processes them with a shared deformable-attention aggregator, and injects the two outputs into the current state:

$$H_t = \Phi_{bista} \left(\left[H_t^{rec}, \Phi_{loc} \left([H_t^{past}, H_t^{future}, F_t] \right) \right] \right), \quad (5)$$

where H_t^{past} and H_t^{future} denote the aggregated local features from the two temporal directions, $\Phi_{loc}(\cdot)$ fuses them with the current-frame feature, and $\Phi_{bista}(\cdot)$ injects the fused evidence into the recurrent state. Separating the two directions preserves temporal order before final fusion, while parameter sharing keeps the module compact.

Within each direction, each support feature is first coarsely warped to the target frame using adjacent or composed flow. Deformable attention then refines this initial correspondence: OAR-refined features are used for query-key matching and offset prediction, while the correspondingly aligned non-OAR support features are kept for value propagation. For each reference-support pair, the offset predictor takes the refined reference feature, the coarsely aligned refined support feature, their difference, and absolute difference as input, and predicts h groups of K residual sampling offsets. Although coarse alignment and residual refinement are applied sequentially, the effective correspondence combines coarse flow guidance with learnable residual correction in the same LR feature coordinate system.

For a temporal group $\mathcal{G} \in \{\text{past}, \text{future}\}$, let $\mathcal{I}_{\mathcal{G}}$ be its support-frame indices. At location $\mathbf{p}$ and head m , the flow-and-offset corrected sampling position is denoted as

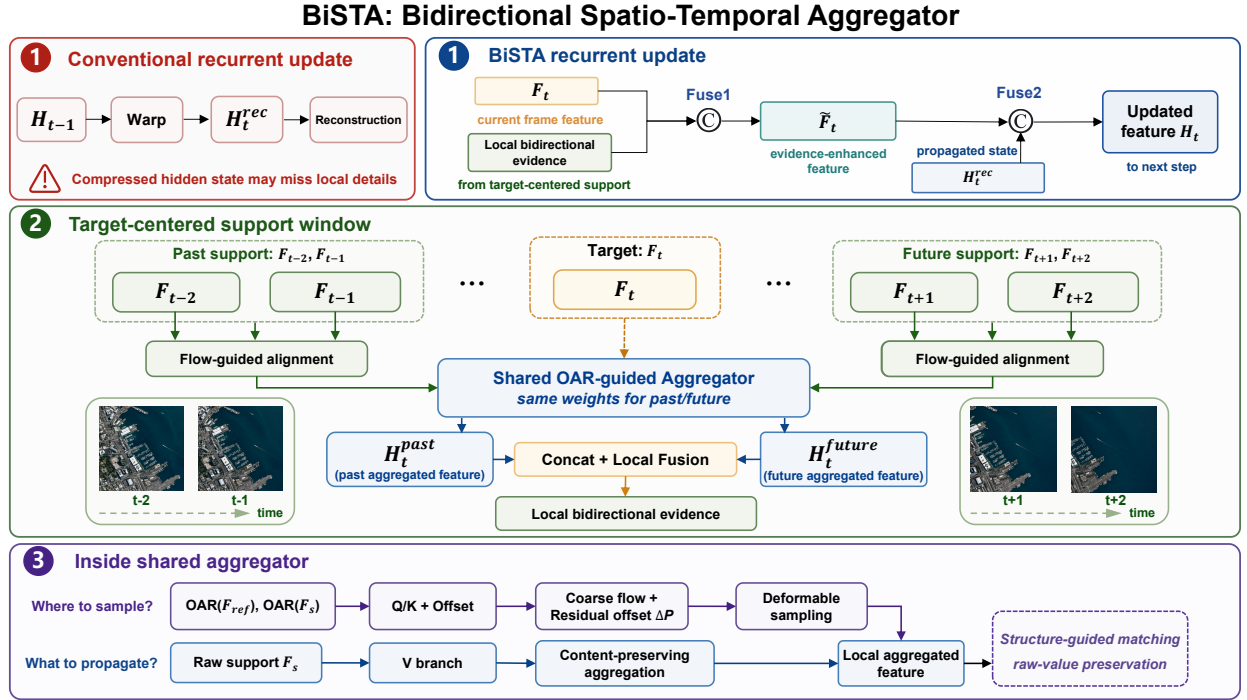


Fig. 4: Structure of the BiSTA module. Local support frames are divided into past and future groups, coarsely aligned by optical flow, refined through deformable attention, and fused with the recurrent state to provide target-centered temporal evidence.

$\mathbf{p}_{m,i,r}$, where $i \in \mathcal{I}_G$ and $r \in \{1, \dots, K\}$. The sampled query, key, and value tokens are defined as

$$\begin{aligned} q_m &= W_m^Q \tilde{F}_{ref}(\mathbf{p}), \\ k_{m,i,r} &= \mathcal{S}(W_m^K \tilde{F}_s^i, \mathbf{p}_{m,i,r}), \\ \mathbf{v}_{m,i,r} &= \mathcal{S}(W_m^V F_s^i, \mathbf{p}_{m,i,r}), \end{aligned} \quad (6)$$

where $\mathcal{S}(\cdot)$ uses bilinear sampling with border padding. The query and key tokens are computed from OAR-refined features for structure-aware matching, while the value token is sampled from the coarsely aligned non-OAR support feature to preserve its original detail content. The group-level attention is then computed by

$$\begin{aligned} \mathbf{o}_m^G(\mathbf{p}) &= \sum_{i \in \mathcal{I}_G} \sum_{r=1}^K A_{m,i,r}(\mathbf{p}) \mathbf{v}_{m,i,r}, \\ A_{m,i,r}(\mathbf{p}) &= \text{Softmax}_{i,r} \left(q_m^\top k_{m,i,r} / \sqrt{d_k} + B_{rel} \right). \end{aligned} \quad (7)$$

where the softmax is applied over all support frames and sampling points in the same temporal group, and B_{rel} provides relative spatio-temporal bias. Outputs from all heads are concatenated and projected to obtain H_t^G . Since the past/future groups share parameters and the local radius is small, the cost remains bounded and does not grow with the full video length.

D. Offset-Aware Refinement

To provide stable structural guidance for local aggregation, we introduce OAR, as shown in Fig. 5. Given an input feature $X \in \mathbb{R}^{C \times H \times W}$, OAR extracts complementary

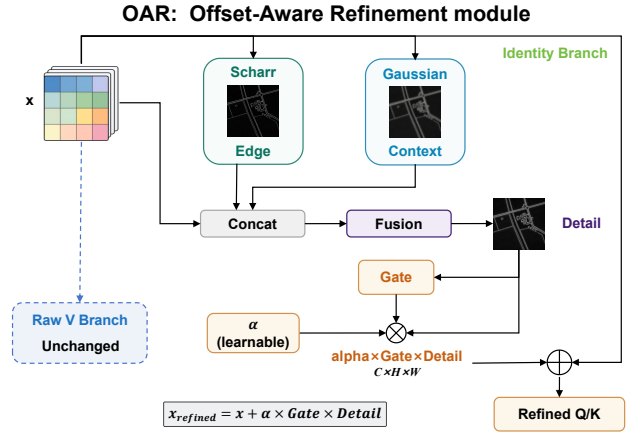


Fig. 5: Structure of OAR. It refines query-key and offset guidance features while preserving raw value features.

identity, edge, and context cues. Two fixed depthwise Scharr filters compute horizontal and vertical gradient responses, while a lightweight learnable context filter initialized with a Gaussian kernel captures local contextual support. Denoting the resulting edge and context responses by E and C_{ctx} , respectively, these cues emphasize structures such as buildings, roads, shorelines, and ship contours. The fixed Scharr operators add no trainable parameters, while the context, fusion, gating, and scaling layers remain lightweight and adaptive.

The original feature, edge prior, and context prior are concatenated and fused to produce a refined guidance

PHASE: Parallel High-frequency Aware Structure Enhancement

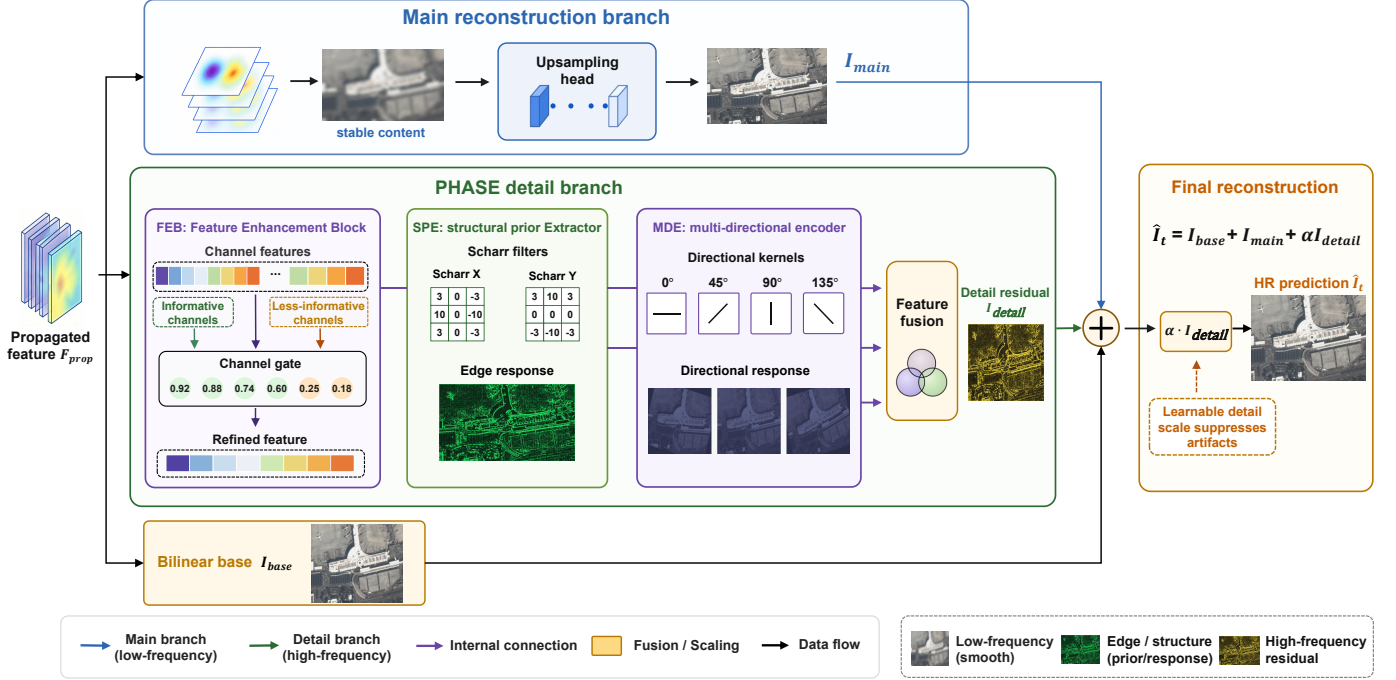


Fig. 6: Structure of PHASE. FEB, SPE, and MDE are fused to generate detail features for the auxiliary reconstruction branch.

representation

$$D = \Phi_{\text{oar}}([X, E, C_{\text{ctx}}]), \quad (8)$$

where $\Phi_{\text{oar}}(\cdot)$ denotes a lightweight learnable fusion function. OAR further predicts a channel-wise gating vector $\mathbf{g} \in [0, 1]^C$ and updates the input feature through residual calibration:

$$X_{\text{out}} = X + \alpha \odot \mathbf{g} \odot D, \quad (9)$$

where α is a learnable channel-wise scaling vector, and $\odot$ denotes element-wise multiplication.

For the reference-support pair in BiSTA, we apply OAR to obtain refined guidance features for query-key construction and offset prediction:

$$\tilde{F}_{\text{ref}} = \text{OAR}(F_{\text{ref}}), \quad \tilde{F}_s^i = \text{OAR}(F_s^i). \quad (10)$$

In contrast, the value branch is still sampled from the original support feature F_s^i , so that OAR improves structural matching and offset estimation without directly perturbing the propagated detail content.

E. High-frequency Dual Upsampling with PHASE

To alleviate over-smoothing in structure-sensitive regions, we adopt a dual-branch reconstruction strategy and introduce PHASE to explicitly compensate for high-frequency details, as shown in Fig. 6. Given the propagated feature F_{prop} , the main branch restores the principal structures and base content, while the auxiliary branch uses PHASE to generate a detail-oriented residual. The two branches are decoded by independent upsampling heads and are fused only at the final image level, which

prevents the detail branch from perturbing the main reconstruction path.

PHASE first projects the propagated feature with a 1×1 convolution and then enhances it with FEB through a residual connection:

$$F_{\text{base}} = \text{FEB}(\Phi_{\text{pre}}(F_{\text{prop}})) + F_{\text{prop}}. \quad (11)$$

The enhanced base feature is then processed by three complementary paths:

$$\begin{aligned} F_{\text{feb}} &= F_{\text{base}}, \\ F_{\text{spe}} &= \text{SPE}(F_{\text{base}}), \\ F_{\text{mde}} &= \text{MDE}(F_{\text{base}}), \end{aligned} \quad (12)$$

where F_{feb} is the adaptively refined feature, F_{spe} is a fixed structural-prior response, and F_{mde} is a learnable multi-directional response. These features are concatenated and fused to produce the residual detail feature:

$$F_{\text{hf}} = \Phi_{\text{fuse}}([F_{\text{feb}}, F_{\text{spe}}, F_{\text{mde}}]), \quad (13)$$

FEB is the learnable refinement path in PHASE. It first normalizes and gates channel responses to separate informative and less-informative features. Instead of discarding weak responses, FEB cross-fuses the two groups so that complementary information can flow across channel partitions. It then applies compressed channel aggregation with grouped and point-wise convolutions, followed by global channel recalibration. This design allows FEB to emphasize discriminative high-frequency channels with moderate computation.

SPE and MDE provide complementary structural cues to FEB. SPE uses fixed depthwise Scharr filters to extract

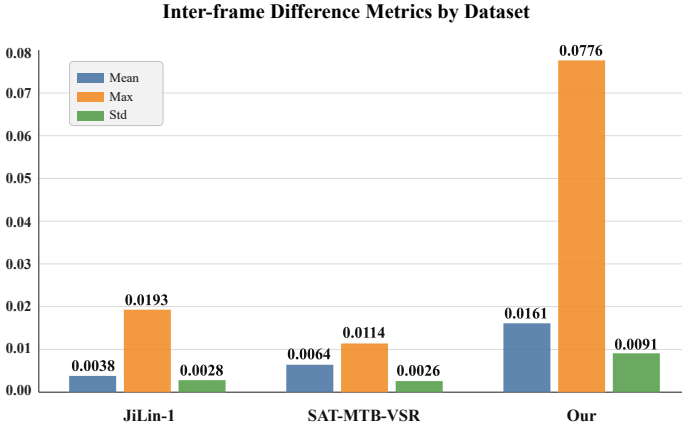


Fig. 7: Inter-frame difference statistics on the evaluation split of JILIN-1, SAT-MTB-VSR, and CQ1-VSR. CQ1-VSR exhibits larger motion magnitude and stronger frame-to-frame variation.

horizontal and vertical gradients and computes their magnitude, providing stable edge-sensitive responses without adding trainable parameters. MDE first reduces the feature dimension and then applies learnable depthwise directional kernels initialized from rotated Sobel operators at 0° , 45° , 90° , and 135° before a 1×1 fusion layer. This branch captures anisotropic structures common in remote sensing scenes, such as roads, shorelines, ship contours, and building arrays. Thus, the bilinear base preserves coarse geometry, the main branch reconstructs principal structures, and PHASE contributes a high-frequency residual whose final strength is controlled by the learnable α .

The auxiliary detail branch provides an explicit pathway for high-frequency residual learning, allowing the main branch to focus on stable reconstruction. The learned detail scale α globally controls the strength of the auxiliary residual and reduces the risk of over-amplifying high-frequency responses. Spatial adaptivity is provided by the PHASE-generated detail feature through feature enhancement, structural-prior extraction, and directional encoding. The effectiveness of this design is examined by the PHASE component and detail-scaling ablations.

IV. Experiments

A. Datasets

We evaluate BiSTAVSR on SAT-MTB-VSR [5] and our CQ1-VSR dataset. SAT-MTB-VSR provides a standardized public benchmark, while CQ1-VSR is collected from original QK-1 satellite videos and retains larger platform-induced inter-frame motion. The two datasets therefore play complementary roles: SAT-MTB-VSR supports comparison with a public RSISR benchmark, and CQ1-VSR stresses temporal aggregation under stronger real satellite-video motion.

Figure 7 compares inter-frame difference distributions on the evaluation splits of JILIN-1, SAT-MTB-VSR, and

CQ1-VSR, while Table IV provides full-dataset GT statistics for SAT-MTB-VSR and CQ1-VSR. Both analyses show that CQ1-VSR exhibits stronger temporal variation. Over the full dataset, CQ1-VSR has substantially larger flow magnitudes and raw GT inter-frame MAE than SAT-MTB-VSR. After motion compensation, its GT residual is reduced by 61.65%, indicating that the additional variation mainly arises from geometric displacement rather than random appearance fluctuation.

For SAT-MTB-VSR, we follow the official split with 413 training and 18 validation sequences, each containing 100 frames at 640×640 resolution. CQ1-VSR contains 19 real satellite scenes, 213 clips, and 15036 image pairs before filtering, covering urban buildings, road networks, coastal regions, and other structure-rich landscapes, as illustrated in Fig. 8. Clips dominated by extensive cloud cover or nearly textureless open-ocean regions are removed because they contain little recoverable structural information. To avoid scene leakage, clips from the same source scene are assigned to only one subset, rather than randomly splitting adjacent clips or frames. The final train/validation/test subsets contain 10464, 2202, and 1314 LR-HR image pairs, respectively.

CQ1-VSR uses real satellite videos as HR references and generates LR inputs through a reproducible synthetic degradation pipeline, since accurately paired real LR-HR satellite videos are difficult to acquire. Thus, the “real” property of CQ1-VSR refers to the source scenes and their temporal motion, while the supervised LR-HR pairs are produced by a documented degradation process. All HR frames are processed at 640×640 , and LR frames are generated with an overall $\times 4$ degradation adapted from blind real-world SR protocols [37]–[39] and related degradation modeling studies [40], [41]. For each clip, one degradation configuration is sampled and applied to all frames to avoid artificial frame-wise flicker. The first stage applies a randomly sampled blur kernel followed by area downsampling. The second stage performs JPEG compression, applies an optional secondary blur, and then conducts the second downsampling step. Finally, the output is resized to exactly one quarter of the HR resolution. The blur kernels include sinc, Gaussian, generalized Gaussian, and plateau kernels with random anisotropy and rotation, while the two downsampling scales are chosen jointly to match the target $\times 4$ factor. The full degradation script and detailed parameter ranges will be released with the dataset.

For evaluation, we adopt PSNR and SSIM [42] as distortion-oriented metrics and LPIPS [43] as a perception-oriented metric. These complementary metrics reflect the common fidelity-perception tradeoff in image restoration [44]. Unless otherwise specified, quantitative results are reported on the luminance channel.

B. Implementation Details

Our method is implemented in PyTorch for $\times 4$ RSISR. During training, the model takes 15 consecutive LR

TABLE I: Quantitative comparison on SAT-MTB-VSR and CQ1-VSR. Params, FLOPs, and Time are reported in M, G, and ms, respectively. FLOPs and runtime are measured under the temporal setting listed in the Frames column. Best and second-best results are highlighted for clarity.

Method	Venue	Frames	Params	FLOPs	Time	SAT-MTB-VSR			CQ1-VSR		
						PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Bicubic	/	1	/	/	/	37.1314	0.9422	0.2047	35.7739	0.9251	0.4600
EDVR [12]	CVPR 2019	5	20.7	897.07	35.36	39.4158	0.9625	0.1370	40.2581	0.9537	0.3379
BasicVSR [6]	CVPR 2021	15	6.3	2480.93	144.07	39.8688	0.9663	0.1383	40.8427	0.9596	0.3091
IconVSR [6]	CVPR 2021	15	8.7	3115.61	201.35	39.8721	0.9647	0.1431	40.8429	0.9603	0.3048
BasicVSR++ [7]	CVPR 2022	15	7.3	2719.39	203.13	39.6378	0.9652	0.1415	41.1048	0.9610	0.2996
RVRT [8]	NeurIPS 2022	16	10.8	4362.17	1147.28	39.9596	0.9661	0.1386	41.1355	0.9615	0.3012
RASVSR [5]	JSTARS 2023	15	5.4	1975.21	145.64	40.0866	0.9672	0.1347	41.1532	0.9611	0.3028
VRT [13]	TIP 2024	16	30.7	8117.63	1700.94	40.3669	0.9681	0.1339	40.5266	0.9576	0.3225
MIA [14]	CVPR 2024	16	16.6	8618.89	3818.22	39.1511	0.9606	0.1543	41.2675	0.9619	0.2960
IA-RT [15]	CVPR 2024	16	13.4	13925.83	4623.71	39.0331	0.9594	0.1538	40.9804	0.9609	0.3054
MADNet [10]	TIP 2025	15	9.2	3293.88	248.75	39.7631	0.9644	0.1406	41.1638	0.9615	0.2953
NSBRNet [24]	TCSVT 2026	15	5.87	/	/	37.58	0.9508	/	/	/	/
BiSTAVSR (ours)	/	15	12.6	5242.01	455.49	40.7814	0.9713	0.1214	41.4101	0.9629	0.2951
BasicVSR++ [7]	CVPR 2022	30	7.3	5499.74	408.08	40.9969	0.9728	0.1217	41.3467	0.9624	0.2911
RASVSR [5]	JSTARS 2023	30	5.4	3989.09	289.95	41.1550	0.9728	0.1346	41.3391	0.9624	0.2974
NSBRNet [24]	TCSVT 2026	30	5.87	/	/	38.75	0.9599	/	/	/	/
BiSTAVSR (ours)	/	30	12.6	10384.44	904.71	41.2607	0.9736	0.1123	41.5266	0.9632	0.2902

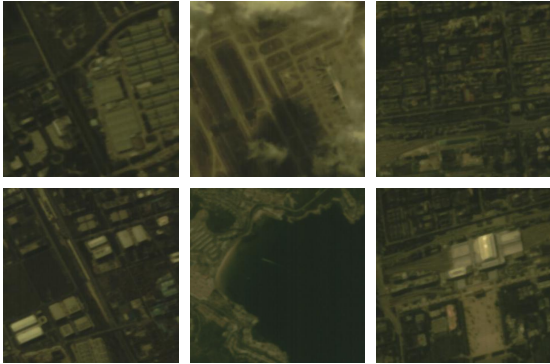


Fig. 8: Example frames from CQ1-VSR. The dataset contains diverse real satellite scenes with large temporal variation and rich structural details.

frames as input. Optical flow is estimated by pretrained SPyNet [26]. Apart from SPyNet, all remaining components, including the retained MADE and EDVR keyframe branches and the newly introduced BiSTA, OAR, PHASE, and reconstruction modules, are initialized and trained from scratch. EDVR-based keyframe features are extracted at a fixed stride and fused into the propagation state.

All reproduced models are trained separately on SAT-MTB-VSR and CQ1-VSR using the corresponding splits. We use 4 NVIDIA H200 GPUs with a total batch size of 4. LR patches of size 64×64 are cropped from 15-frame clips, with random horizontal and vertical flipping for augmentation. The optimizer is Adam with an initial learning rate of 2×10^{-4} , updated by CosineAnnealingRestartLR for 100,000 iterations. We use the Charbonnier loss as the reconstruction objective.

For fair comparison, baselines with accessible implementations are retrained with the same LR-HR pairs,

splits, batch size, and training iterations. Official or commonly used BasicSR implementations are adopted when available, and only the dataset loader and crop size are adapted. NSBRNet is treated separately because a usable implementation was unavailable at the time of evaluation. We therefore quote its published 15- and 30-frame PSNR/SSIM results on SAT-MTB-VSR as a literature reference, while CQ1-VSR, FLOPs, runtime, and LPIPS results are not reported. During inference, the temporal length follows the Frames column in Table I; this keeps each reproduced method under its official or commonly used temporal setting instead of forcing a single input length that may be unsupported by some architectures. FLOPs and runtime are measured under the listed temporal setting and include SPyNet for flow-guided methods. PSNR and SSIM are reported on the luminance channel, and LPIPS is used as a perceptual metric.

C. Comparison with State-of-the-Art Methods

We compare BiSTAVSR with a broad set of representative video super-resolution methods, including both general VSR models and approaches tailored to remote sensing or real-world degradations, such as EDVR, BasicVSR, IconVSR, BasicVSR++, RVRT, RASVSR, VRT, MIA, IA-RT, and MADNet. Among these competitors, MADNet is particularly relevant to our study because it is specifically designed for remote sensing video super-resolution and already provides a strong spatial-frequency modeling baseline. All baselines in Table I are retrained on the corresponding dataset rather than directly transferred from natural-video checkpoints.

Table I summarizes the quantitative comparisons on SAT-MTB-VSR and CQ1-VSR. BiSTAVSR gives the best overall restoration quality among the compared methods.

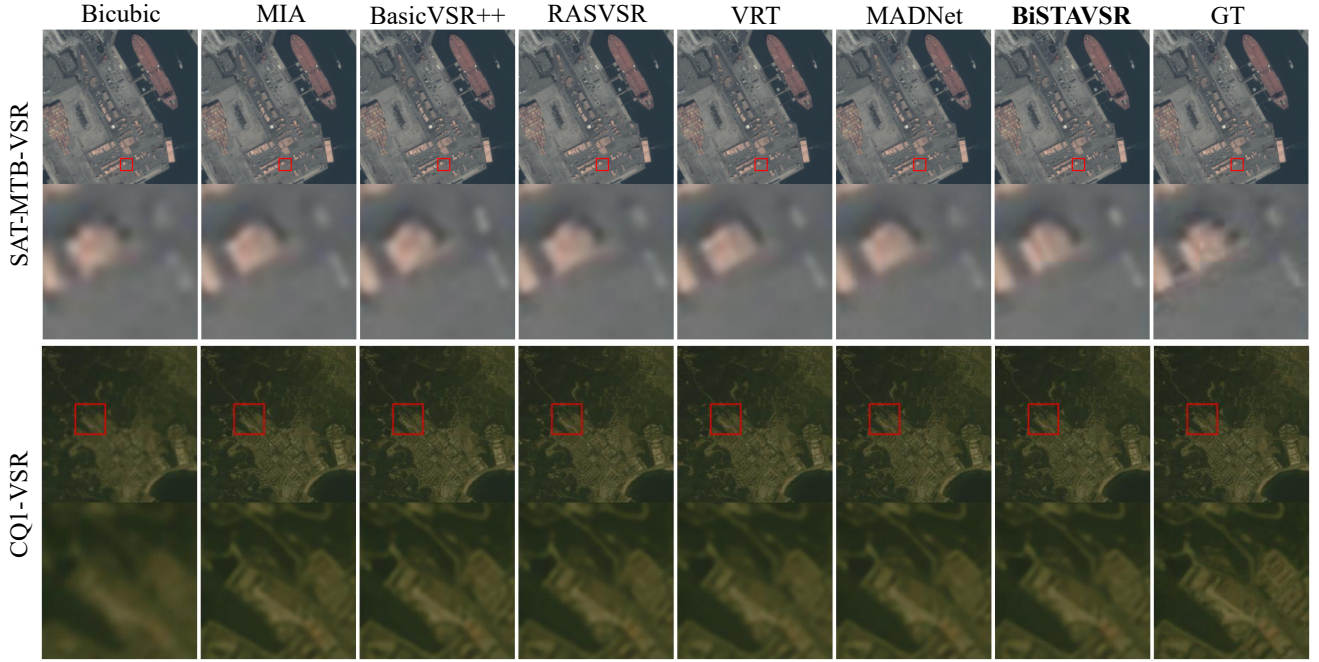


Fig. 9: Qualitative comparison on SAT-MTB-VSR and CQ1-VSR. BiSTAVSR recovers clearer structures and finer textures, especially in challenging regions.

TABLE II: Overall module ablation and efficiency analysis. Params and runtime are reported in M and ms, respectively.

Method	BiSTA	OAR	PHASE	Params	Time	SAT-MTB-VSR		CQ1-VSR	
						PSNR(dB) $\uparrow$	SSIM $\uparrow$	PSNR(dB) $\uparrow$	SSIM $\uparrow$
MADNet baseline				9.19	248.78	39.7631	0.9644	41.1638	0.9615
+ PHASE			$\checkmark$	9.70	295.02	40.1672	0.9690	41.2881	0.9621
+ BiSTA	$\checkmark$			12.03	409.68	40.2405	0.9680	41.2752	0.9620
+ BiSTA + OAR	$\checkmark$	$\checkmark$		12.06	413.39	40.4852	0.9698	41.3088	0.9623
+ BiSTA + PHASE	$\checkmark$		$\checkmark$	12.54	456.53	40.4899	0.9701	41.3734	0.9624
+ BiSTA + OAR + PHASE	$\checkmark$	$\checkmark$	$\checkmark$	12.56	458.62	40.7814	0.9713	41.4101	0.9629

Relative to MADNet, it improves PSNR from 39.7631 to 40.7814 on SAT-MTB-VSR and from 41.1638 to 41.4101 on CQ1-VSR under the primary 15-frame setting. The same trend on CQ1-VSR is noteworthy because this dataset contains larger temporal variation and more realistic scene-scale motion than the public benchmark.

On SAT-MTB-VSR, the proposed method improves all three quality metrics, indicating that the gain is not limited to a single distortion measure. The PSNR and SSIM improvements suggest that recurrent propagation combined with local BiSTA aggregation can reconstruct more accurate frame-level structures. The LPIPS reduction further indicates that the restored frames are perceptually closer to the reference, which is consistent with the role of PHASE in enhancing high-frequency details. Since SAT-MTB-VSR is relatively standardized and has been widely used for comparison, these results demonstrate that the proposed modules do not overfit to the newly introduced CQ1-VSR setting. For methods that provide a long-sequence recurrent inference setting, the lower block of Table I further compares the corresponding 30-frame versions. BiSTAVSR still achieves the best results under

this longer temporal setting, showing that the proposed local aggregation and detail enhancement remain effective when more frames are available.

On CQ1-VSR, the performance gap among methods more directly reflects their ability to handle stronger satellite-video motion and richer temporal variation. Window-based methods such as EDVR benefit from target-centered local support, while recurrent and transformer-based methods exploit longer temporal contexts through propagation or spatio-temporal attention. Nevertheless, these strategies either rely on a fixed local window or compress temporal information into propagated states, which can limit their ability to use reliable neighboring evidence under large scene-scale shifts. BiSTAVSR achieves the best PSNR, SSIM, and LPIPS, suggesting that recurrent memory, local past/future BiSTA support, OAR-guided matching, and PHASE detail recovery are complementary for realistic satellite video sequences.

To connect the main comparison with the contribution of each component, Table II reports a compact module-level ablation together with parameter and runtime changes. BiSTA brings the main gain and cost by

TABLE III: Temporal consistency comparison on SAT-MTB-VSR and CQ1-VSR. FDE denotes frame-difference error, and warping error is computed after motion compensation. Both metrics are mean absolute RGB intensity errors reported on the 8-bit scale. Lower values are better.

Method	SAT FDE (8-bit) ↓	SAT Warp. Err. (8-bit) ↓	CQ1 FDE (8-bit) ↓	CQ1 Warp. Err. (8-bit) ↓
Bicubic	1.4852	1.4214	3.0261	0.9097
EDVR [12]	1.3840	1.1913	3.0667	0.8499
BasicVSR [6]	1.3333	1.0595	2.2585	0.7781
IconVSR [6]	1.3190	1.0379	2.2313	0.7785
BasicVSR++ [7]	1.3086	1.0322	2.2155	0.7493
RVRT [8]	1.3073	1.0687	3.0665	0.8525
RASVSR [5]	1.3078	1.0592	2.1844	0.7373
VRT [13]	1.3200	1.1079	2.3101	0.9388
MIA [14]	1.3804	1.1623	2.1689	0.8204
IA-RT [15]	1.4923	1.4193	2.1899	0.7723
MADNet [10]	1.3343	1.0570	2.1838	0.7474
BiSTAVSR (ours)	1.2487	1.0340	2.1225	0.7316

TABLE IV: Full-dataset temporal change statistics under the GT protocol. Flow magnitudes are estimated on adjacent GT frames and reported in HR pixels. GT MAE values are normalized RGB errors in $[0, 1]$.

Metric	SAT-MTB-VSR	CQ1-VSR	Ratio
Flow mean (HR px)	0.2155	8.0112	37.18×
Flow P95 (HR px)	0.3867	9.6800	25.03×
Raw GT MAE ($[0, 1]$)	0.00612	0.01383	2.26×
Warp GT MAE ($[0, 1]$)	0.00592	0.00530	—
Warp reduction (%)	3.41%	61.65%	—

adding explicit local temporal aggregation, OAR adds only a small overhead while improving offset guidance, and PHASE further improves reconstruction through high-frequency compensation.

D. Quantitative and Qualitative Results

The joint improvements in PSNR, SSIM, and LPIPS indicate that BiSTAVSR increases reconstruction fidelity without relying on overly smooth outputs.

Figure 9 shows representative visual results on both SAT-MTB-VSR and CQ1-VSR. In challenging regions such as building boundaries, road networks, and port areas, several competing methods exhibit local blur, texture loss, or motion-related misalignment. By comparison, BiSTAVSR produces cleaner boundaries and finer local structures. This behavior is also visible on CQ1-VSR under stronger real-source temporal variation and clip-consistent synthetic degradation.

Figure 10 visualizes reconstruction error maps on both datasets. The darker responses of BiSTAVSR around building boundaries, road-like structures, and small objects indicate lower local reconstruction errors than MADNet.

E. Temporal Consistency Analysis

Frame-wise metrics do not fully characterize video restoration quality. A method may produce visually sharp individual frames while still introducing temporal flicker

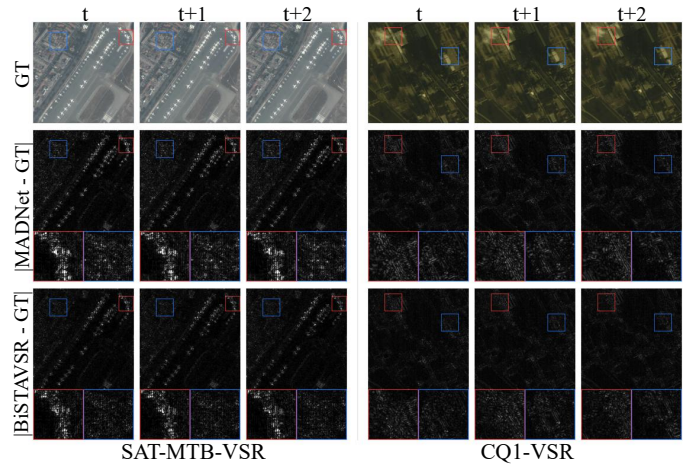


Fig. 10: Visual comparison of reconstruction error maps on SAT-MTB-VSR and CQ1-VSR. The error maps are computed as the absolute difference between the restored results and the GT frames. Brighter pixels denote larger reconstruction errors. BiSTAVSR shows reduced errors in structure-sensitive regions with MADNet.

or unstable edges across consecutive frames. This issue is particularly important for satellite videos because many downstream applications rely on consistent object boundaries and stable scene structures over time. BiSTAVSR is expected to improve temporal consistency through two mechanisms: recurrent propagation provides long-range memory, while BiSTA injects local bidirectional support around the target frame and reduces dependence on a single warped hidden state.

FDE is computed as the mean absolute difference between the SR and GT frame differences, i.e., $|(I_t^{SR} - I_{t-1}^{SR}) - (I_t^{GT} - I_{t-1}^{GT})|$. For warping error, pretrained SPyNet estimates backward flow from the current GT frame to the previous GT frame; the previous SR frame is warped to the current grid and compared with the current SR frame over valid in-frame samples. Both RGB errors are converted to the 8-bit scale and averaged over all adjacent frame pairs.

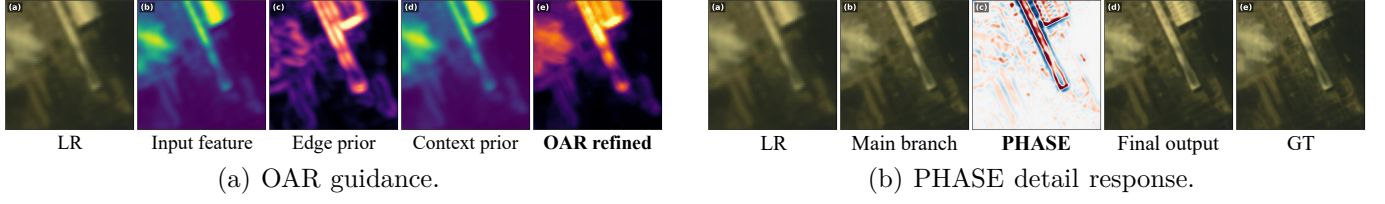


Fig. 11: Mechanism visualizations of OAR and PHASE. OAR combines fixed edge priors with learned context guidance for boundary-aware matching, while PHASE concentrates the auxiliary residual on high-frequency structures.

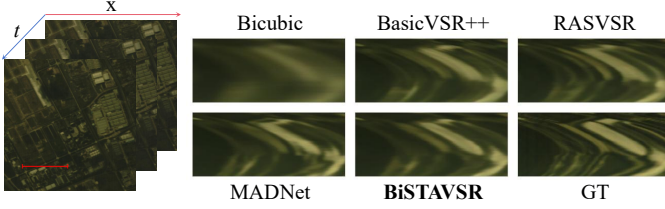


Fig. 12: Qualitative comparison of temporal profiles on CQ1-VSR. A fixed image row is stacked across the timeline to reveal temporal changes. Clearer and more continuous profile structures indicate more stable restoration.

TABLE V: Comparison of neighbor-frame aggregation strategies on CQ1-VSR under the same recurrent framework.

Model	Aggregation	PSNR(dB) $\uparrow$	SSIM $\uparrow$
MADNet	none	41.1638	0.9615
	DCN	41.2034	0.9618
	EDVR	40.9584	0.9603
	BiSTA	41.2752	0.9620

As reported in Table III, BiSTAVSR obtains the lowest FDE on both datasets and the lowest warping error on CQ1-VSR. On SAT-MTB-VSR, its warping error is close to the best value while providing substantially higher PSNR, SSIM, and LPIPS than the competing methods. These results indicate that the improvement is not obtained at the cost of unstable frame-to-frame behavior.

Figure 12 provides a temporal-profile visualization on CQ1-VSR. Instead of displaying many consecutive frames, we sample the same image row from each restored frame and stack the rows along the temporal axis. This compact view makes temporal discontinuities easier to observe: unstable restoration often produces broken or mixed streak patterns, while temporally consistent reconstruction preserves smoother and more continuous structures.

F. Complexity and Efficiency Analysis

The quality-efficiency tradeoff is important for RSVSR because long satellite videos can contain many frames and may need to be processed under operational time constraints. BiSTAVSR introduces BiSTA, OAR, and PHASE, but it controls the added cost in several ways. BiSTA uses a small local radius and shared parameters for past and future aggregation. OAR combines fixed

TABLE VI: Ablation study of structural priors in OAR on CQ1-VSR.

Model	Prior setting	PSNR(dB) $\uparrow$	SSIM $\uparrow$
BiSTAVSR	w/o OAR	41.3734	0.9624
	no fixed prior	41.3919	0.9626
	Sobel	41.3669	0.9623
	Laplacian	41.3888	0.9627
	learnable prior	41.3619	0.9622
	Scharr	41.4101	0.9629

TABLE VII: Ablation study of OAR branch design on CQ1-VSR. Evaluated on MADNet+BiSTA.

Identity	Edge	Context	PSNR(dB) $\uparrow$	SSIM $\uparrow$
✓	✓	✓	41.3088	0.9623
✓		✓	41.1653	0.9615
	✓	✓	41.2569	0.9621
✓	✓		41.2289	0.9619
			41.2752	0.9620

depthwise Scharr operators with a lightweight Gaussian-initialized context filter, fusion, and channel gating. PHASE is placed in an auxiliary residual branch with a controllable detail scale.

Fig. 1 provides a parameter-PSNR comparison, while Table I reports FLOPs and runtime under the listed temporal setting of each method. Compared with heavier transformer-based methods, BiSTAVSR obtains higher restoration accuracy with substantially lower runtime than VRT, MIA, and IA-RT. Although it is more expensive than lightweight recurrent baselines, the additional cost is associated with explicit local aggregation and high-frequency enhancement, which bring consistent gains in both distortion-oriented and perceptual metrics.

G. Ablation Studies

Fine-grained design ablations are conducted on CQ1-VSR because its stronger temporal variation makes changes in aggregation, offset guidance, and detail reconstruction easier to distinguish.

To further examine the aggregation design, Table V compares BiSTA with alternative neighbor-frame aggregation strategies on CQ1-VSR. BiSTA achieves the best performance, confirming that the proposed bidirectional local aggregation is more effective than directly inserting

TABLE VIII: Ablation study of OAR injection strategy on CQ1-VSR. QK denotes query-key/offset guidance, and QKV additionally refines value features.

Model	Injection	PSNR(dB) $\uparrow$	SSIM $\uparrow$
MADNet+BiSTA	w/o OAR	41.2752	0.9620
	QK	41.3088	0.9623
	QKV	41.2178	0.9614

TABLE IX: Ablation study of PHASE branch design on CQ1-VSR.

Main(MADNet)	SPE	MDE	FEB	PSNR(dB) $\uparrow$	SSIM $\uparrow$
✓	✓	✓	✓	41.2881	0.9621
✓	✓		✓	41.2741	0.9619
✓		✓	✓	41.1972	0.9613
✓	✓	✓		41.1254	0.9613
✓				41.1638	0.9615

conventional local aggregation modules into the recurrent framework.

The comparison also clarifies why BiSTA is not simply another local alignment block. DCN improves sampling flexibility but lacks explicit attention-based selection among sampled support features. EDVR-style aggregation provides strong local support, but when inserted into a recurrent pipeline it may not match the temporal symmetry required by bidirectional propagation. BiSTA uses shared past/future aggregation and fuses the result with the recurrent state, making it more compatible with long-sequence propagation.

We further examine the internal designs of OAR and PHASE. Tables VI–VIII show that Scharr is the best structural prior, the identity/edge/context inputs are complementary, and QK-only injection is preferable because it guides matching without altering the raw value stream. For PHASE, Table IX shows that combining SPE, MDE, and FEB gives the best branch design. The response visualizations in Fig. 11 further show that OAR strengthens boundary-sensitive guidance, while PHASE concentrates its auxiliary residual around high-frequency structures.

1) Additional Ablations: Table X shows that $k = 2$ gives the best balance between useful local evidence and alignment reliability. The learnable detail scale in Table XI also outperforms no-detail and fixed-weight fusion, indicating that adaptive detail compensation better controls artifacts.

The ablation results are consistent with the visual comparisons in Fig. 9, where BiSTAVSR preserves sharper and more continuous structures in building boundaries, road networks, and other detail-sensitive regions. In particular, the PHASE ablation in Table IX supports the benefit of explicit high-frequency modeling, while the overall gains on CQ1-VSR in Table II indicate that the proposed design remains effective under stronger real-source temporal variation.

TABLE X: Ablation study of the local temporal radius k in BiSTA on CQ1-VSR.

Radius k	Support frames	PSNR(dB) $\uparrow$	SSIM $\uparrow$
0	none	41.2881	0.9621
1	$t \pm 1$	41.3826	0.9623
2	$t \pm 1, t \pm 2$	41.4101	0.9629
3	$t \pm 1, t \pm 2, t \pm 3$	41.3973	0.9625

TABLE XI: Ablation study of the PHASE detail scaling parameter α on CQ1-VSR.

Model	Strategy	PSNR(dB) $\uparrow$	SSIM $\uparrow$
BiSTAVSR	No detail branch	41.3088	0.9623
	Fixed α	41.3448	0.9625
	Learnable α	41.4101	0.9629

V. Conclusion

This paper presented BiSTAVSR, an offset-aware bidirectional spatio-temporal aggregation framework for remote sensing video super-resolution. The method is designed for satellite videos where neighboring frames provide useful high-resolution evidence but also contain larger platform-induced motion and structure-sensitive alignment errors. BiSTAVSR keeps the efficiency of recurrent video propagation while inserting BiSTA for explicit local past/future aggregation, uses OAR to guide offset prediction and query-key matching with structural priors, and introduces PHASE to compensate high-frequency residual details through a separate auxiliary branch.

In addition, CQ1-VSR was constructed from original QK-1 videos to complement existing RSVSR benchmarks with stronger inter-frame variation and richer real satellite motion. It uses real HR satellite source sequences and a reproducible clip-consistent degradation pipeline to generate supervised LR inputs. Experiments on SAT-MTB-VSR and CQ1-VSR show that BiSTAVSR achieves favorable reconstruction quality, perceptual quality, temporal alignment metrics, and parameter efficiency. The ablation studies further verify the contributions of BiSTA, OAR, PHASE, local temporal radius, structural priors, and adaptive detail scaling, indicating the importance of domain-adapted spatio-temporal aggregation and structure-aware detail restoration for satellite video reconstruction.

References

- [1] S. A. Ahmadi, A. Ghorbanian, and A. Mohammadzadeh, “Moving vehicle detection, tracking and traffic parameter estimation from a satellite video: A perspective on a smarter city,” *International journal of remote sensing*, vol. 40, no. 22, pp. 8379–8394, 2019.
- [2] Y. Gu, T. Wang, X. Jin, and G. Gao, “Detection of event of interest for satellite video understanding,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 11, pp. 7860–7871, 2020.
- [3] J. Shao, B. Du, C. Wu, M. Gong, and T. Liu, “Hrsiam: High-resolution siamese network, towards space-borne satellite video tracking,” *IEEE Transactions on Image Processing*, vol. 30, pp. 3056–3068, 2021.

- [4] Y. Xiao, X. Su, Q. Yuan, D. Liu, H. Shen, and L. Zhang, "Satellite video super-resolution via multiscale deformable convolution alignment and temporal grouping projection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–19, 2021.
- [5] H. Wang, S. Li, and M. Zhao, "A lightweight recurrent aggregation network for satellite video super-resolution," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 17, pp. 685–695, 2023.
- [6] K. C. Chan, X. Wang, K. Yu, C. Dong, and C. C. Loy, "Basicvrs: The search for essential components in video super-resolution and beyond," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 4947–4956.
- [7] K. C. Chan, S. Zhou, X. Xu, and C. C. Loy, "Basicvrs++: Improving video super-resolution with enhanced propagation and alignment," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 5972–5981.
- [8] J. Liang, Y. Fan, X. Xiang, R. Ranjan, E. Ilg, S. Green, J. Cao, K. Zhang, R. Timofte, and L. V. Gool, "Recurrent video restoration transformer with guided deformable attention," *Advances in Neural Information Processing Systems*, vol. 35, pp. 378–393, 2022.
- [9] Y. Xiao, Q. Yuan, K. Jiang, X. Jin, J. He, L. Zhang, and C.-W. Lin, "Local-global temporal difference learning for satellite video super-resolution," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 34, no. 4, pp. 2789–2802, 2024.
- [10] Y. Xiao, Q. Yuan, K. Jiang, Y. Chen, S. Wang, and C.-W. Lin, "Multi-axis feature diversity enhancement for remote sensing video super-resolution," *IEEE Transactions on Image Processing*, 2025.
- [11] S. Pang, J. Zeng, P. Qin, and G. Qiao, "Ebrnet: Lightweight enhanced bidirectional recurrent network for satellite video super-resolution," *IEEE Transactions on Geoscience and Remote Sensing*, 2026.
- [12] X. Wang, K. C. Chan, K. Yu, C. Dong, and C. Change Loy, "Edvr: Video restoration with enhanced deformable convolutional networks," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*, 2019, pp. 0–0.
- [13] J. Liang, J. Cao, Y. Fan, K. Zhang, R. Ranjan, Y. Li, R. Timofte, and L. Van Gool, "Vrt: A video restoration transformer," *IEEE Transactions on Image Processing*, vol. 33, pp. 2171–2182, 2024.
- [14] X. Zhou, L. Zhang, X. Zhao, K. Wang, L. Li, and S. Gu, "Video super-resolution transformer with masked inter&intra-frame attention," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 25399–25408.
- [15] K. Xu, Z. Yu, X. Wang, M. B. Mi, and A. Yao, "Enhancing video super-resolution via implicit resampling-based alignment," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 2546–2555.
- [16] D. Zhang, J. Shao, Z. Liang, X. Liu, and H. T. Shen, "Multi-branch networks for video super-resolution with dynamic reconstruction strategy," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 31, no. 10, pp. 3954–3966, 2021.
- [17] Y. Luo, L. Zhou, S. Wang, and Z. Wang, "Video satellite imagery super resolution via convolutional neural networks," *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 12, pp. 2398–2402, 2017.
- [18] A. Xiao, Z. Wang, L. Wang, and Y. Ren, "Super-resolution for 'jilin-1' satellite video imagery via a convolutional network," *Sensors*, vol. 18, no. 4, p. 1194, 2018.
- [19] K. Jiang, Z. Wang, P. Yi, and J. Jiang, "A progressively enhanced network for video satellite imagery superresolution," *IEEE Signal Processing Letters*, vol. 25, no. 11, pp. 1630–1634, 2018.
- [20] H. Liu, Y. Gu, T. Wang, and S. Li, "Satellite video super-resolution based on adaptively spatiotemporal neighbors and nonlocal similarity regularization," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 12, pp. 8372–8383, 2020.
- [21] H. Liu and Y. Gu, "Deep joint estimation network for satellite video super-resolution with multiple degradations," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–15, 2022.
- [22] Z. He and D. He, "A unified network for arbitrary scale super-resolution of video satellite images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 10, pp. 8812–8825, 2021.
- [23] Z. He, D. He, X. Li, and J. Xu, "Unsupervised video satellite super-resolution by using only a single video," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, p. 3040972, 2022.
- [24] Y. Wang, Z. Shao, X. Zuo, T. Lu, J. Wang, Y. Wang, S. Wang, Z. Zhang, and X. Zhao, "Nsbrnet: Non-local spatio-temporal bidirectional recurrent network for satellite video super-resolution," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 36, no. 2, pp. 1473–1486, 2026.
- [25] N. Ni and L. Zhang, "Deformable convolution alignment and dynamic scale-aware network for continuous-scale satellite video super-resolution," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–17, 2024.
- [26] A. Ranjan and M. J. Black, "Optical flow estimation using a spatial pyramid network," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 4161–4170.
- [27] Z. Teed and J. Deng, "Raft: Recurrent all-pairs field transforms for optical flow," in *European conference on computer vision*. Springer, 2020, pp. 402–419.
- [28] Y. Zhang, H. Wang, H. Zhu, and Z. Chen, "Optical flow reusing for high-efficiency space-time video super resolution," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 33, no. 5, pp. 2116–2128, 2023.
- [29] J. Dai, H. Qi, Y. Xiong, Y. Li, G. Zhang, H. Hu, and Y. Wei, "Deformable convolutional networks," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 764–773.
- [30] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable detr: Deformable transformers for end-to-end object detection," *arXiv preprint arXiv:2010.04159*, 2020.
- [31] Z. Qin, P. Zhang, F. Wu, and X. Li, "Fcanet: Frequency channel attention networks," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 783–792.
- [32] L. Yang, R.-Y. Zhang, L. Li, and X. Xie, "Simam: A simple, parameter-free attention module for convolutional neural networks," in *International Conference on Machine Learning*. PMLR, 2021, pp. 11863–11874.
- [33] X. Mao, Y. Liu, F. Liu, Q. Li, W. Shen, and Y. Wang, "Intriguing findings of frequency selection for image deblurring," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 2, 2023, pp. 1905–1913.
- [34] F. Li, L. Zhang, Z. Liu, J. Lei, and Z. Li, "Multi-frequency representation enhancement with privilege information for video super-resolution," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 12814–12825.
- [35] X. Lang, J. Zhang, T. Lu, Y. Yao, Y. Wang, and L. Wang, "Frequency-aware enhancement network for satellite video super-resolution," *Remote Sensing*, vol. 17, no. 24, p. 3994, 2025.
- [36] T. Zhang, H. Chen, S. Chen, and C. Bian, "Edge-enhanced efficient network for remote sensing image super-resolution," *International journal of remote sensing*, vol. 43, no. 14, pp. 5324–5347, 2022.
- [37] K. Zhang, J. Liang, L. Van Gool, and R. Timofte, "Designing a practical degradation model for deep blind image super-resolution," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 4791–4800.
- [38] X. Wang, L. Xie, C. Dong, and Y. Shan, "Real-esrgan: Training real-world blind super-resolution with pure synthetic data," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 1905–1914.
- [39] Y. Hao and F. Yu, "Super-resolution degradation model: Converting high-resolution datasets to optical zoom datasets," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 33, no. 11, pp. 6374–6389, 2023.
- [40] L. Wang, Y. Wang, X. Dong, Q. Xu, J. Yang, W. An, and Y. Guo, "Unsupervised degradation representation learning for blind super-resolution," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 10581–10590.
- [41] Z. Luo, Y. Huang, S. Li, L. Wang, and T. Tan, "Learning the degradation distribution for blind image super-resolution," *arXiv preprint arXiv:2203.04962*, 2022.
- [42] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural

similarity,” *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.

- [43] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, “The unreasonable effectiveness of deep features as a perceptual metric,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 586–595.
- [44] Y. Blau and T. Michaeli, “The perception-distortion tradeoff,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 6228–6237.



Senwei Hu is currently working toward his M.S. degree at the Hangzhou Institute of Advanced Study, University of Chinese Academy of Sciences, Hangzhou, China. His research interests include image and video super-resolution.



Yisheng Zhao received the Ph.D. degree from the College of Information Science and Electronic Engineering, Zhejiang University, Hangzhou, China, in 2024. From 2024 to 2026, he was a Postdoctoral Researcher with the Hangzhou Institute of Advanced Study, University of Chinese Academy of Sciences, Hangzhou, China. He is currently a faculty member with the Hangzhou Institute of Advanced Study, University of Chinese Academy of Sciences, Hangzhou, China. His research

interests include remote sensing visual-language models, fault tolerance of large models, hallucinations of large models, and super-resolution reconstruction.



Yibo Wang is currently an M.S. student with the Hangzhou Institute of Advanced Study, University of Chinese Academy of Sciences, Hangzhou, China. His research mainly focuses on image and video super-resolution.



Hongyi Chen received the Ph.D. degree from the School of Engineering, University of Edinburgh, Edinburgh, U.K., in 2021. He is currently a Postdoctoral Researcher with the Hangzhou Institute of Advanced Study, University of Chinese Academy of Sciences, Hangzhou, China. His research interests include computer vision, remote sensing image processing, physics-informed machine learning, and edge computing.



Yaoqi Liu is currently an Associate Researcher with the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China. He received the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China. His research interests include computer architecture, space-based computing architecture, space-based large models and intelligent applications, fault-tolerant computing, and the integration of computing, communication, navigation, and remote sensing. He has led or participated in several national and institutional research projects and has published more than 30 papers and filed more than 50 invention patents. He has received several honors, including the President Award of the Chinese Academy of Sciences.



Hao Zhou is currently working toward his Ph.D. degree at the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China. He is also affiliated with the Hangzhou Institute for Advanced Study, University of Chinese Academy of Sciences, Hangzhou, China. His research interests include space-based computing systems and applications.



Yinhe Han is currently a Professor with the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China. He received the Ph.D. degree from the Institute of Computing Technology, Chinese Academy of Sciences, in 2006. He was promoted to Associate Professor in 2008 and Professor in 2014. His research interests include computer architecture and chips, intelligent robotics, intelligent hardware, data computing chips, and computing systems. He has received several honors and awards, including the Second Prize of the National Technological Invention Award in 2012 and the Excellent Young Scientists Fund from the National Natural Science Foundation of China in 2015.



Jianyu Wang is the Dean of the Hangzhou Institute of Advanced Study, University of Chinese Academy of Sciences, Hangzhou, China, and a Chief Professor with the School of Physics and Optoelectronic Engineering. He is also an Academician of the Chinese Academy of Sciences. His research mainly focuses on space optoelectronic technologies and systems. He has led the design and development of the system for the world's first quantum science experimental satellite, addressing key technical challenges such as beam alignment, polarization preservation, and single-photon detection in satellite-to-ground quantum experiments. He has also made significant contributions to hyperspectral remote sensing, laser remote sensing, and high-sensitivity spaceborne laser detection. He has received several national and provincial-level science and technology awards, including two Second Prizes of the National Technological Invention Award.